

30 DAYS OF MACHINE LEARNING

Study Notes — Days 1–10

Fatemeh Babaei

A concise but complete review of the foundational concepts covered during the first ten days of the challenge.

Section	Focus
Days 1–3	Machine Learning foundations and the relationship between AI and ML
Days 4–7	Core learning paradigms and how machines learn from data
Day 8	Train/Test Split and Cross-Validation
Day 9	Self-Supervised Learning
Day 10	Features and Labels

DAY 01 — What Is Machine Learning?

Core Concept

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every individual case.

Key Points

- A traditional program follows explicitly written rules.
- A machine learning model learns a useful relationship from examples.
- The quality and relevance of data strongly affect the learned model.
- The goal is not simply to memorize training examples, but to **generalize** to unseen data.

Example

Example: A spam filter can learn from previously labeled emails and use learned patterns to classify new messages.

IMPORTANT TERMS

• Artificial Intelligence (AI) • Machine Learning (ML) • Model • Training Data • Generalization

DAY 02 — Machine Learning in Everyday Life

Core Concept

Machine Learning is already used in many systems that make predictions, recommendations, classifications, or rankings.

Key Points

- Recommendation systems suggest movies, products, music, or posts.
- Email systems classify messages as spam or non-spam.
- Fraud detection systems identify unusual transactions.
- Search and ranking systems estimate which results are most relevant.
- Speech and image systems learn patterns from large collections of examples.

Example

Key idea: ML is useful when manually writing every rule would be difficult, expensive, or impossible.

IMPORTANT TERMS

• Prediction • Classification • Recommendation • Pattern Recognition • Data

DAY 03 — Artificial Intelligence vs. Machine Learning

Core Concept

Artificial Intelligence is the broader field of building systems that perform tasks associated with intelligent behavior. Machine Learning is one major approach used to achieve AI.

Key Points

- **AI** is the broader concept.

- **ML** is a subset of AI based on learning from data.
- **Deep Learning** is a subset of ML based on multi-layer neural networks.
- Not every AI system must use machine learning, although modern AI relies heavily on ML.

Example

A useful mental model is: **AI** → **Machine Learning** → **Deep Learning**.

IMPORTANT TERMS

• Artificial Intelligence • Machine Learning • Deep Learning • Neural Network

Quick Check: Is every ML system an AI system? Yes. Is every AI system necessarily ML-based? No.

DAY 04 — Learning from Data: The Core Idea

Core Concept

At the center of machine learning is a simple idea: provide data, define what the model should learn, train it, and evaluate whether it works on data it has not seen before.

Key Points

- **Input data** contains information available to the model.
- The model identifies patterns or relationships in that data.
- Training adjusts model parameters to reduce prediction error.
- Evaluation checks whether the learned pattern works beyond the training examples.

Example

The central objective is **generalization**: good performance on new, unseen examples.

IMPORTANT TERMS

• Input • Output • Model • Training • Generalization

DAY 05 — Supervised Learning

Core Concept

Supervised Learning uses examples where the desired output is known. The model learns a mapping from input data to a target output.

Key Points

- Training examples contain both **features** and a known **target/label**.
- The model learns from these labeled examples.
- It can then predict the label for new examples.
- Two common supervised tasks are **classification** and **regression**.

Example

Example: Given house size, location, and number of rooms, predict the house price.

IMPORTANT TERMS

DAY 06 — Unsupervised Learning

Core Concept

Unsupervised Learning works with data where the desired target is not provided. The model searches for useful structure or patterns within the data.

Key Points

- There is no predefined target label for each training example.
- The model may discover groups, similarities, or lower-dimensional representations.
- **Clustering** is a common unsupervised learning task.
- Unsupervised methods are useful for exploration and discovering hidden structure.

Example

Example: Group customers into segments based on their behavior without giving the model predefined customer categories.

IMPORTANT TERMS

• Unsupervised Learning • Clustering • Pattern Discovery • Similarity

DAY 07 — Semi-Supervised Learning

Core Concept

Semi-Supervised Learning combines a small amount of labeled data with a larger amount of unlabeled data.

Key Points

- Labeling data can be expensive or time-consuming.
- Unlabeled data is often much easier to collect.
- The model uses both sources of information during learning.
- This approach can be useful when only a limited portion of the dataset has reliable labels.

Example

Example: In medical imaging, only some images may have expert annotations while thousands of additional images remain unlabeled.

IMPORTANT TERMS

• Semi-Supervised Learning • Labeled Data • Unlabeled Data • Annotation

Quick Check: Why can semi-supervised learning be useful? Because obtaining labels may be expensive while unlabeled data is abundant.

DAY 08 — Train/Test Split and Cross-Validation

Core Concept

A model should not be judged only on the same data used for training. Training and evaluation must be separated so that performance on unseen data can be estimated.

Key Points

- **Training data** is used to fit the model.
- **Test data** is kept separate for a final evaluation.
- A single split can sometimes make the result dependent on one particular partition.
- **Cross-Validation** evaluates the model across multiple folds to obtain a more stable estimate.
- In **5-Fold Cross-Validation**, the training data is divided into five folds; each fold becomes the validation set once while the other four are used for training.

Example

Important distinction: Cross-validation is mainly a model-selection/validation strategy; a separate final test set is still valuable for an unbiased final estimate.

IMPORTANT TERMS

• Train Set • Validation Set • Test Set • Cross-Validation • 5-Fold

Quick Check: Why is a single Train/Test Split sometimes not enough? Because the measured score can depend strongly on one particular split.

DAY 09 — Self-Supervised Learning

Core Concept

Self-Supervised Learning creates a learning signal from the data itself. Instead of requiring humans to manually label every example, the system generates a target from part or structure of the original data.

Key Points

- The data does not need traditional human-provided labels for the pretraining task.
- The model creates a **pseudo-target** or prediction task from the input itself.
- A common example is hiding part of a sequence or sentence and asking the model to predict the missing information.
- The learned representation can later be used for downstream tasks.

Example

Example: Given a sentence such as “Machine Learning is ____”, the model learns to predict a missing word using the surrounding context.

IMPORTANT TERMS

• Self-Supervised Learning • Pseudo-Label • Pretraining • Representation Learning

DAY 10 — Features and Labels

Core Concept

Features are the input variables used by a model to learn. A label is the target outcome the model is trying to predict in a supervised learning problem.

Key Points

- **Feature:** information used by the model as input.
- **Label:** the known target/output associated with an example.
- A dataset may contain multiple features and one target.
- Good feature selection can strongly affect model performance.
- In a house-price problem, size, location, and number of rooms can be features, while price is the label.

Example

A simple mental model: **Features** → **Model** → **Label/Prediction**.

IMPORTANT TERMS

• Feature • Label • Input • Target • Prediction

Quick Check: If the goal is to predict whether an email is spam, what is the label? The spam/non-spam outcome.

DAYS 1–10 — CONSOLIDATED REVIEW

The first ten days build a logical foundation: what Machine Learning is, how it relates to AI, how different learning paradigms use data, how models are evaluated, and how inputs and targets are represented.

Concept	One-line summary
AI	The broad field of building systems that perform intelligent tasks.
Machine Learning	A way for systems to learn patterns from data.
Deep Learning	ML based on multi-layer neural networks.
Supervised Learning	Learns from labeled examples.
Unsupervised Learning	Discovers structure in unlabeled data.
Semi-Supervised Learning	Combines labeled and unlabeled data.
Self-Supervised Learning	Creates learning targets from the data itself.
Feature	An input variable used by a model.
Label / Target	The outcome a supervised model learns to predict.
Train Set	Data used to fit the model.
Validation	Data/process used to compare or tune models during development.
Test Set	Held-out data used for final performance estimation.
Cross-Validation	Repeated validation across multiple folds.

Key Takeaways

- **Data is central:** Machine learning depends on data quality, representation, and the learning objective.
- **Labels change the problem:** supervised learning has known targets; unsupervised learning does not.
- **Not all learning is human-labeled:** self-supervised methods can construct training signals directly from raw data.
- **Evaluation must reflect unseen data:** training performance alone is not enough.
- **Features and labels have different roles:** features provide information; labels represent the target in supervised learning.

Learning Path

Artificial Intelligence → Machine Learning → Learning Paradigms → Data Representation → Training & Validation → Model Evaluation

Prepared for personal study and review — Fatemeh Babaei